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public class BinarySearch {
public static int binarySearch(int[] arr, int target) {
int left = 0;
int right = arr.length - 1;
while (left <= right) {
int mid = left + (right - left) / 2;
if (arr[mid] == target) {
return mid;
}
else if (arr[mid] < target) {
left = mid + 1;
}
else {
right = mid - 1;
}
}
return -1;
}
public static void main(String[] args) {
int[] arr = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
int target = 7;
int result = binarySearch(arr, target);
System.out.println(result != -1 ? "Элемент " + target + " найден на индексе: " + result : "Элемент не найден.");
}
}

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#include <iostream>
#include <vector>
#include <algorithm>

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int binarySearch(const std::vector<int>& arr, int target) {
    int left = 0;
    int right = arr.size() - 1;

    while (left <= right) {
        int mid = left + (right - left) / 2;

        if (arr[mid] == target) {
            return mid;
        }
        else if (arr[mid] < target) {
            left = mid + 1;
        }
        else {
            right = mid - 1;
        }
    }
    return -1;
}

int main() {
    std::vector<int> arr = {1, 2, 3, 4, 5, 6, 7, 8, 9};
    int target = 5;

    int result = binarySearch(arr, target);
    if (result != -1) {
        std::cout << "Элемент найден на индексе: " << result << std::endl;
    } else {
        std::cout << "Элемент не найден" << std::endl;
    }
}

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    return 0;  
}
```